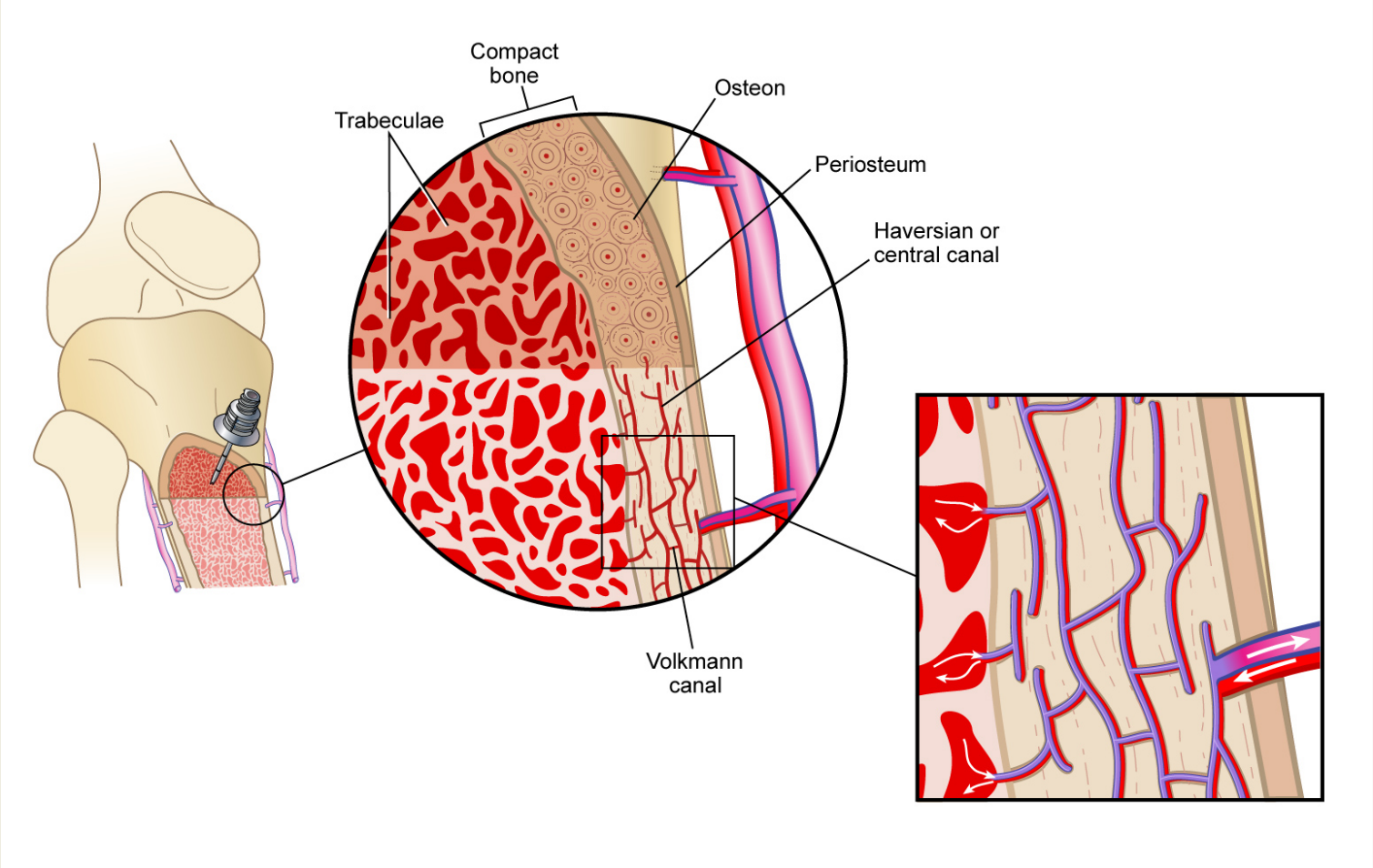
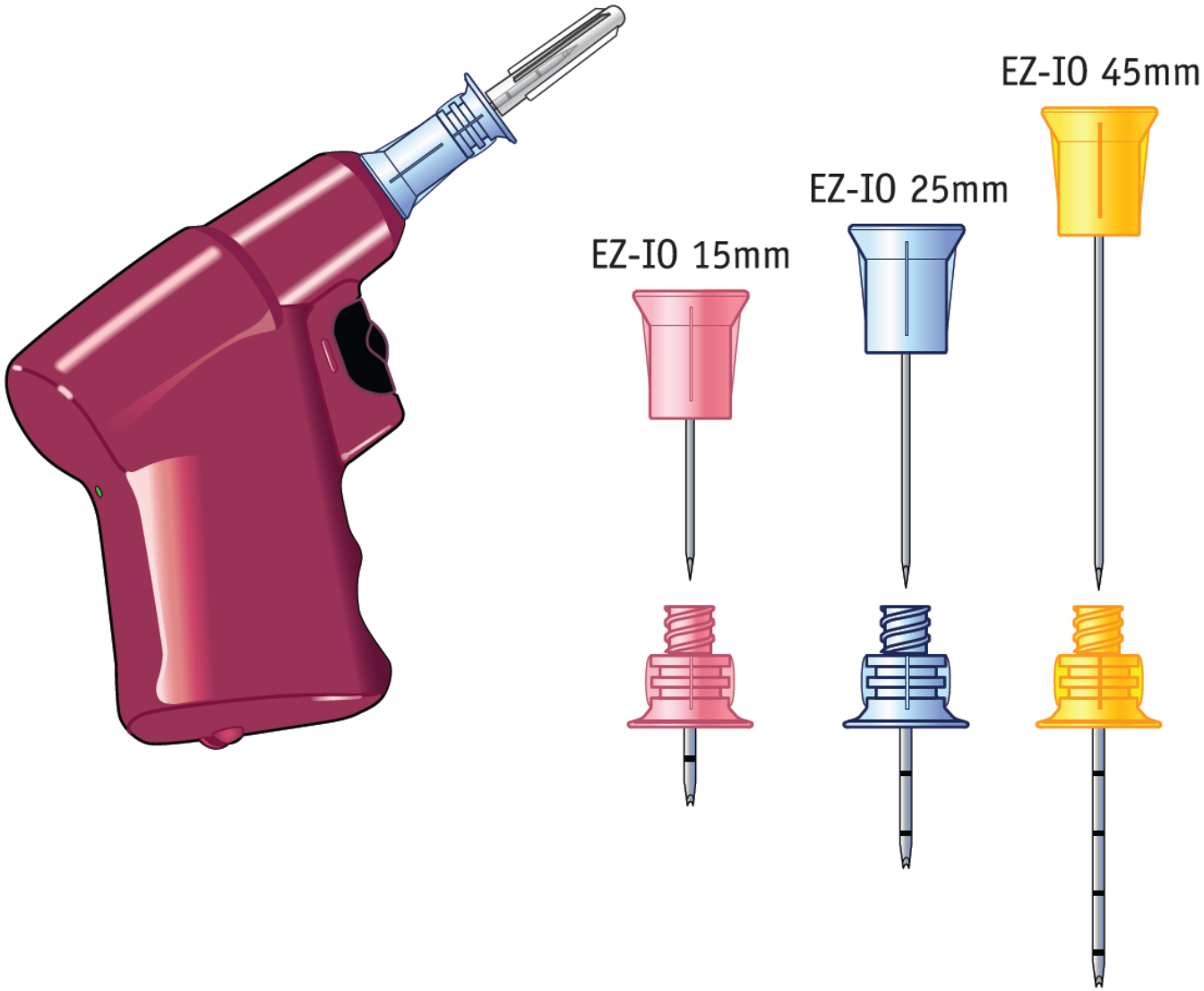
Intraosseous Access

- First discussed in the medical literature in 1922.
- Research and case studiesin the 1940s showed the usefulness of using intraosseous route for administration of blood, fluids and medications.
- An intraosseous device to access the sternum was used during World War II.
- Intraosseous access become a standard in pediatric advanced life support in 1988.

Anatomy of intraosseous access

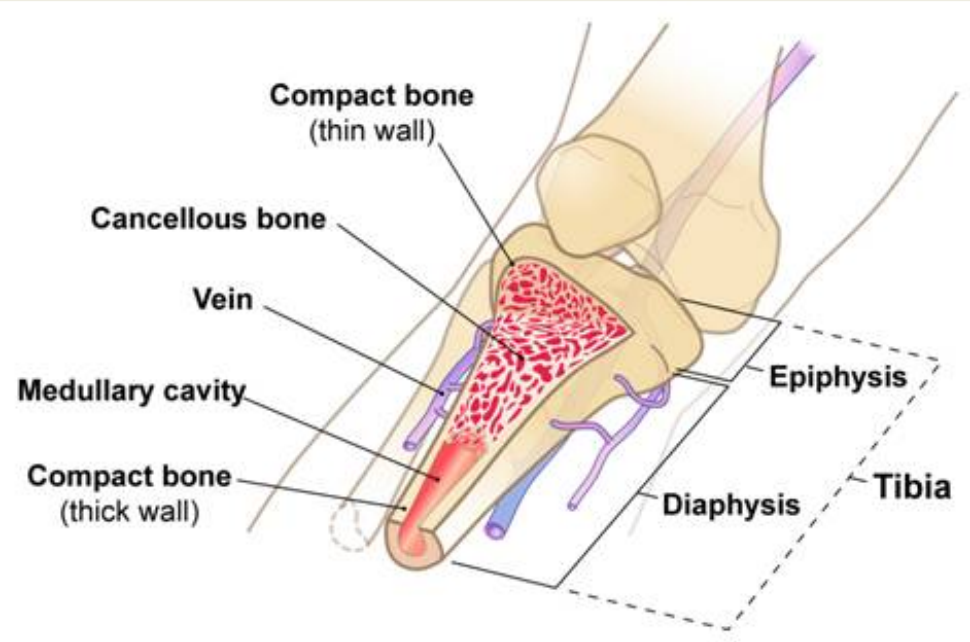


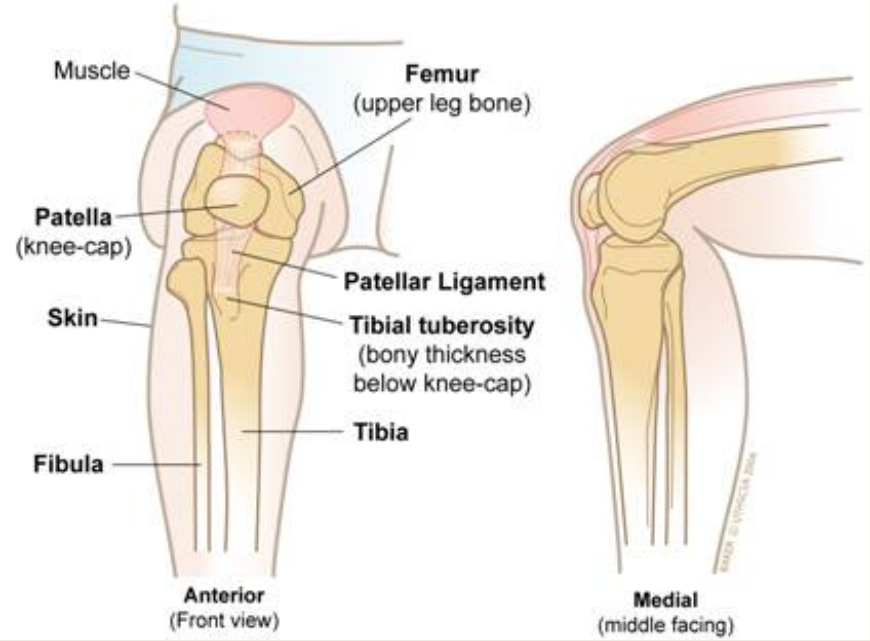
Thousands of small veins lead from the medullary space to the central circulation



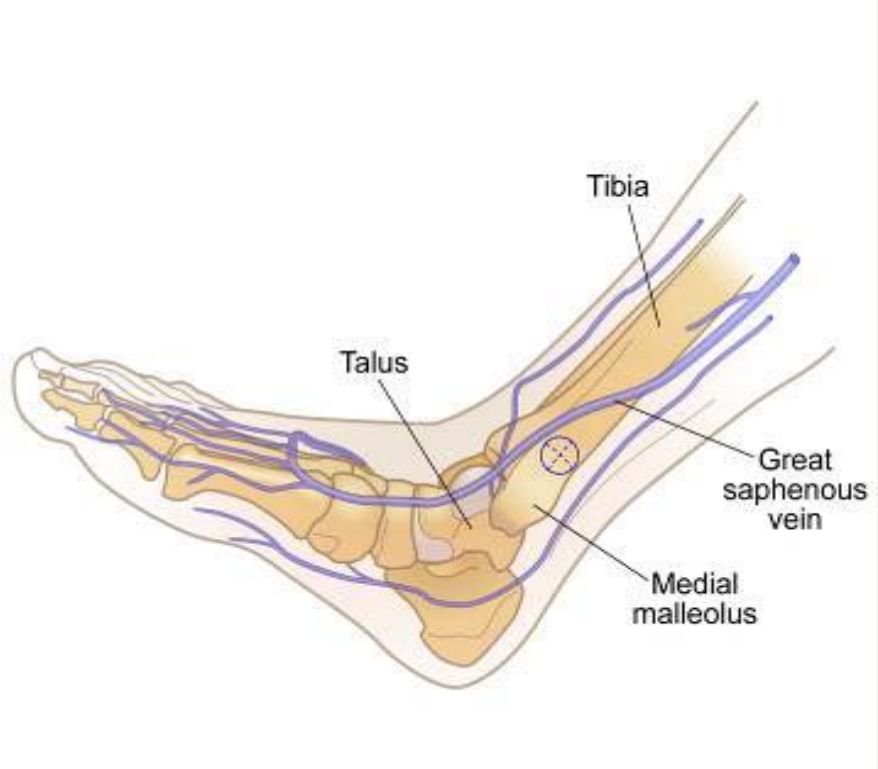
EZ-IO Power Driver and Needle Sets (FIGURE 1)

Proximal Tibial Anatomy

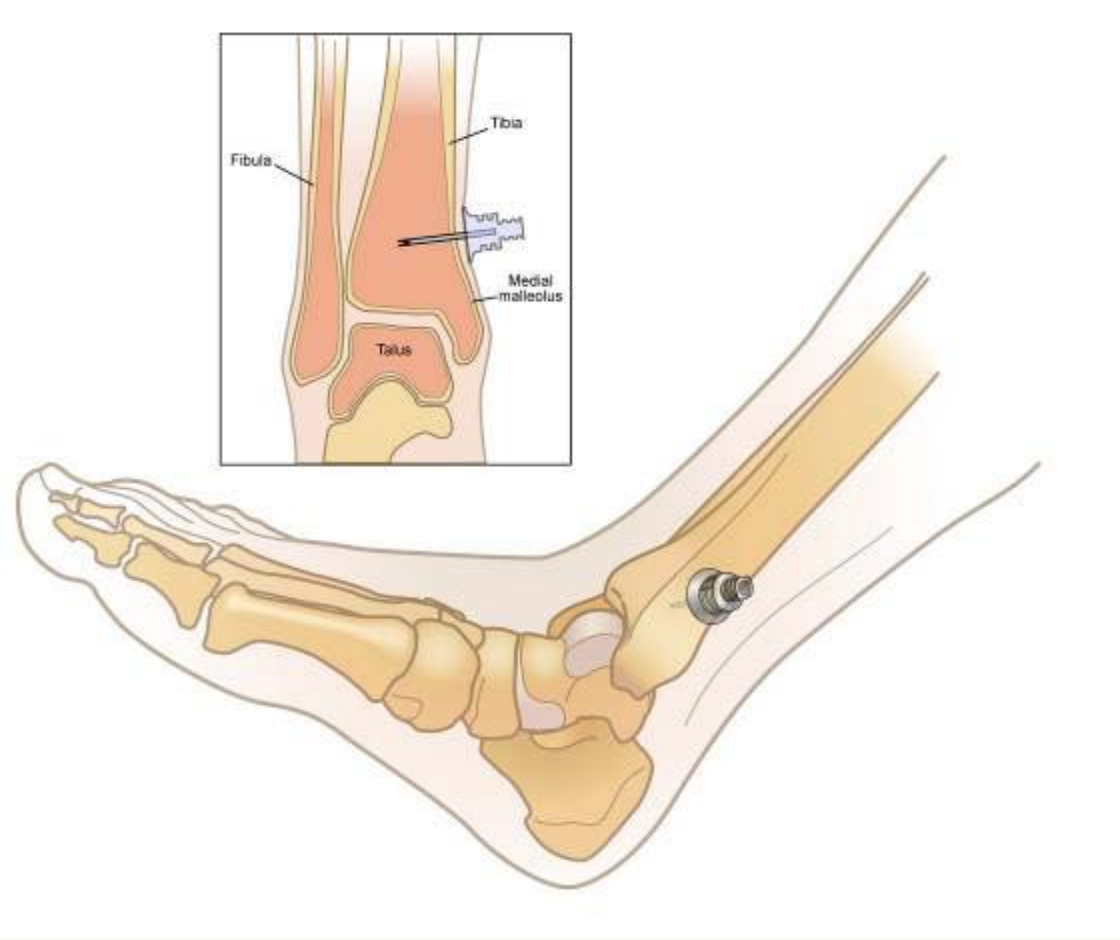




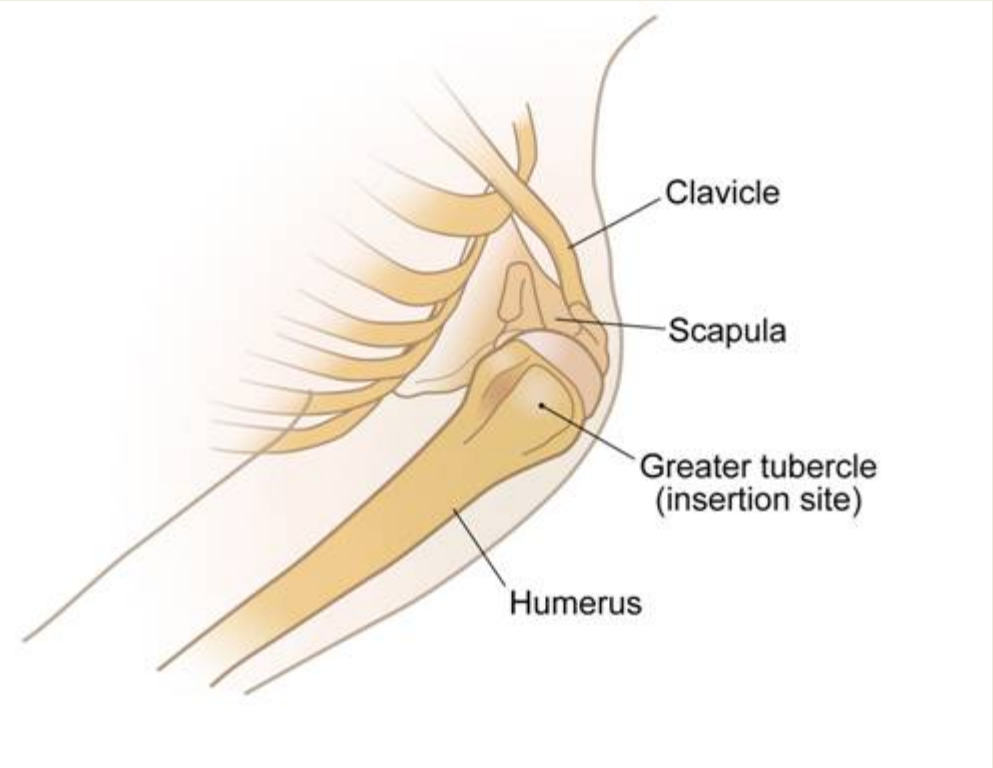
Distal Tibial Anatomy

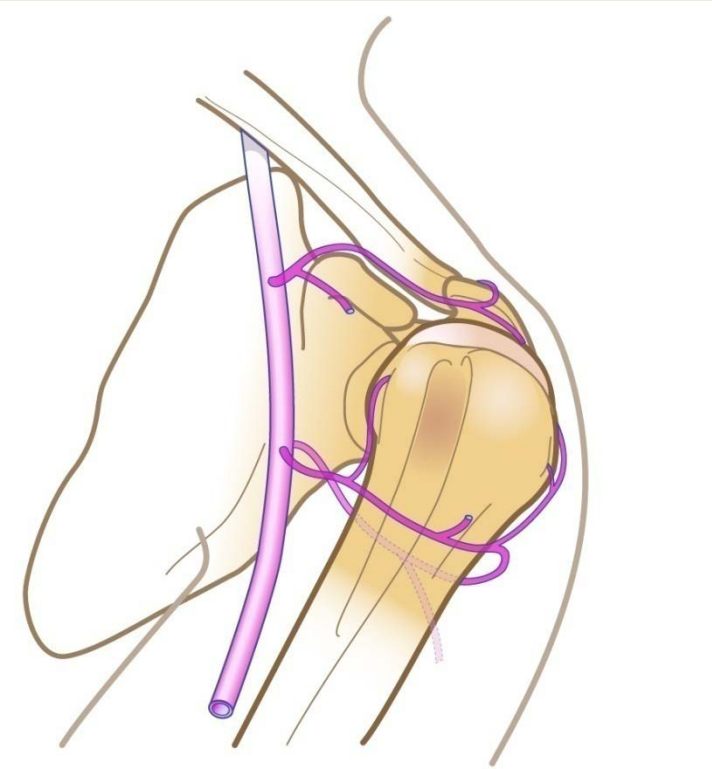


The ankle joint is comprised of the Tibia, Talus and Fibula

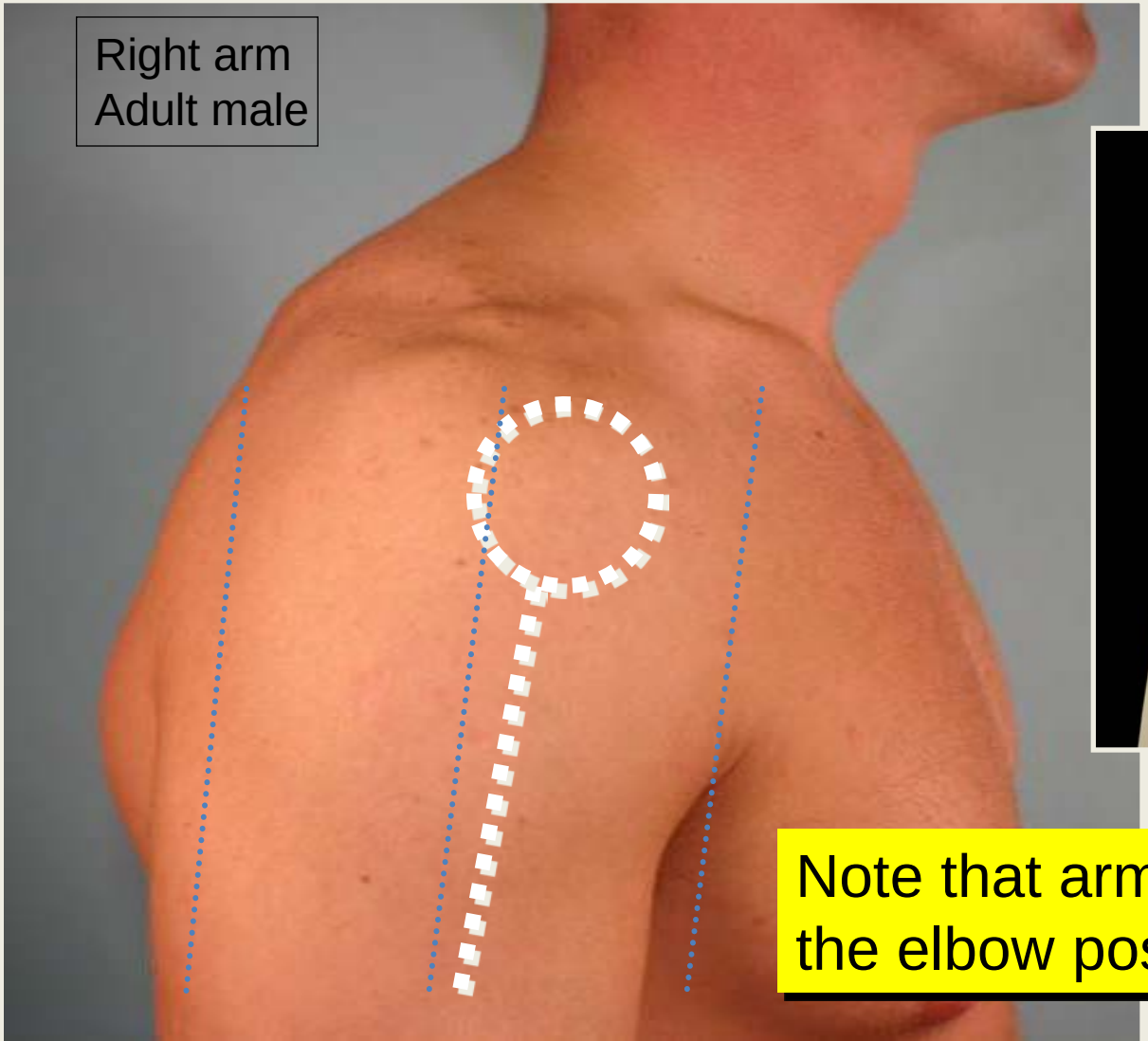


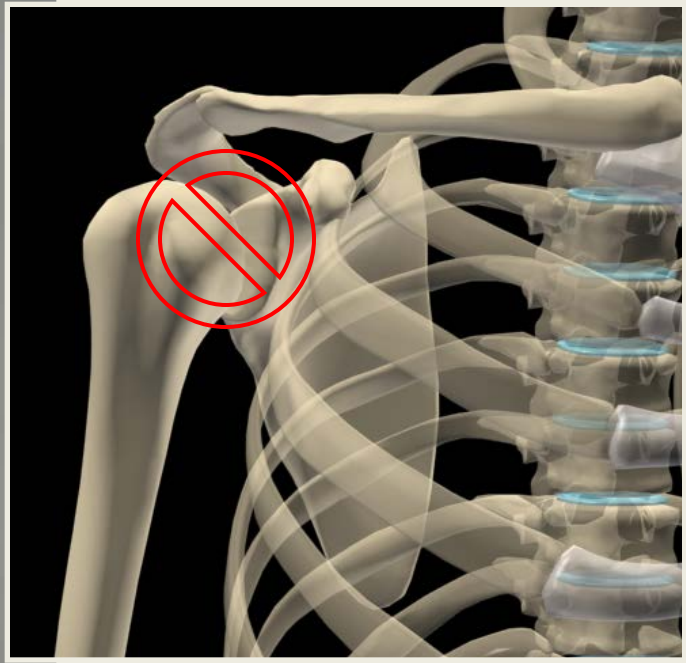
Proximal Humeral Anatomy





The Proximal Humerus insertion site is found “slightly anterior to the arms lateral midline”





Note that arm is adducted with the elbow posteriorly placed!

EZ-IO

Proper Positioning for Use of the Proximal Humerus (Figure 1):

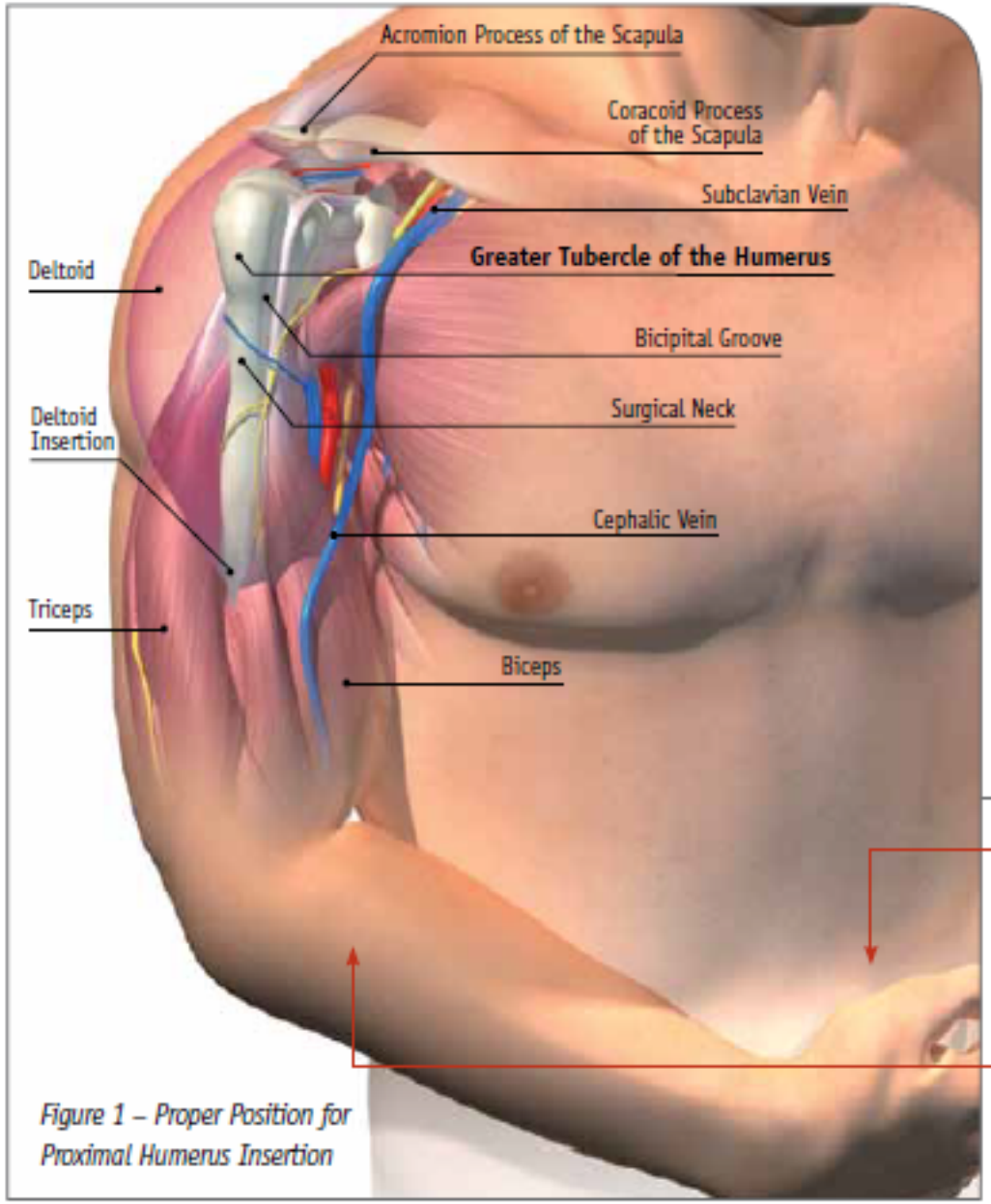


Figure 1 – Proper Position for Proximal Humerus Insertion

The proximal humerus – a relatively new option for intraosseous access – provides additional benefits over the more commonly used tibial locations.[1]

The close proximity of the greater tubercle of the humerus to the heart ensures rapid infusion of medications into the central circulation.[1] Moreover, at least one human study suggests infusion is better tolerated by patients when compared to the tibial sites.[11]

Despite these benefits, the anatomy of the proximal humerus region often discourages its use by clinicians. By becoming familiar with the anatomical structures, clinicians can simplify the site identification process and feel more comfortable accessing the site.

Place the patient's hand over the umbilicus
Causes medial rotation of elbow and humerus

Adduct the arm
Provides greater prominence of insertion site

EZ-IO by vidacare

Steps to Properly Identify the Insertion Site (Figure 2):

Identify the humerus ❶

The humerus is most easily palpated at the insertion point for the deltoid muscle, between the biceps and triceps muscles. This point is approximately mid-way along the length of the arm (see illustration). Palpation of the bone requires firm pressure due to overlying structures.

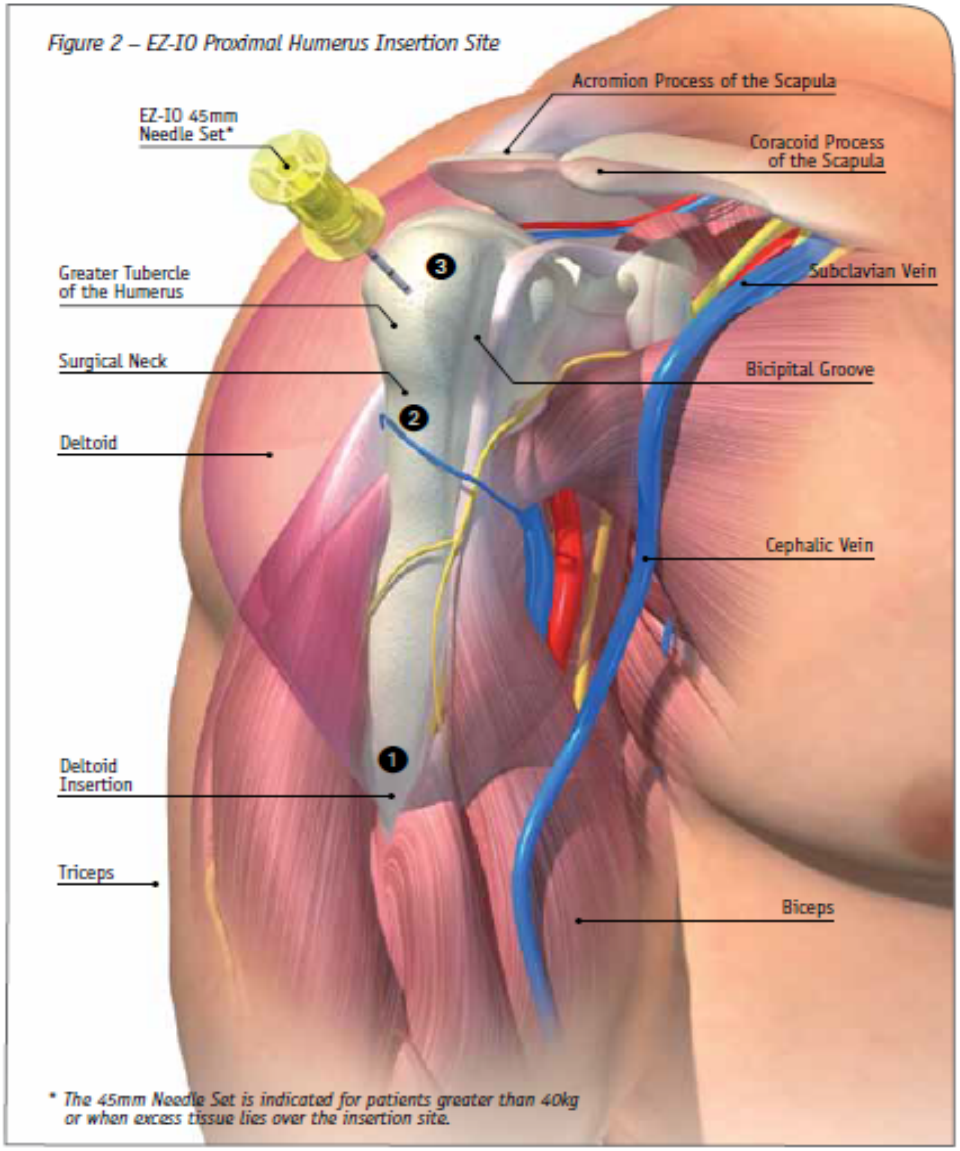
Locate surgical neck of the humerus ❷

The surgical neck can be located by palpating up the length of the humerus until the clinician feels a "notch" or "groove."

Identify insertion site ❸

The appropriate insertion site is approximately 1 cm above the surgical neck for most adults.

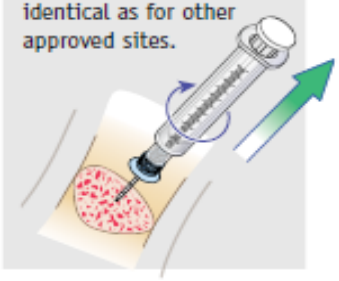
Figure 2 – EZ-IO Proximal Humerus Insertion Site



** The 45mm Needle Set is indicated for patients greater than 40kg or when excess tissue lies over the insertion site.*

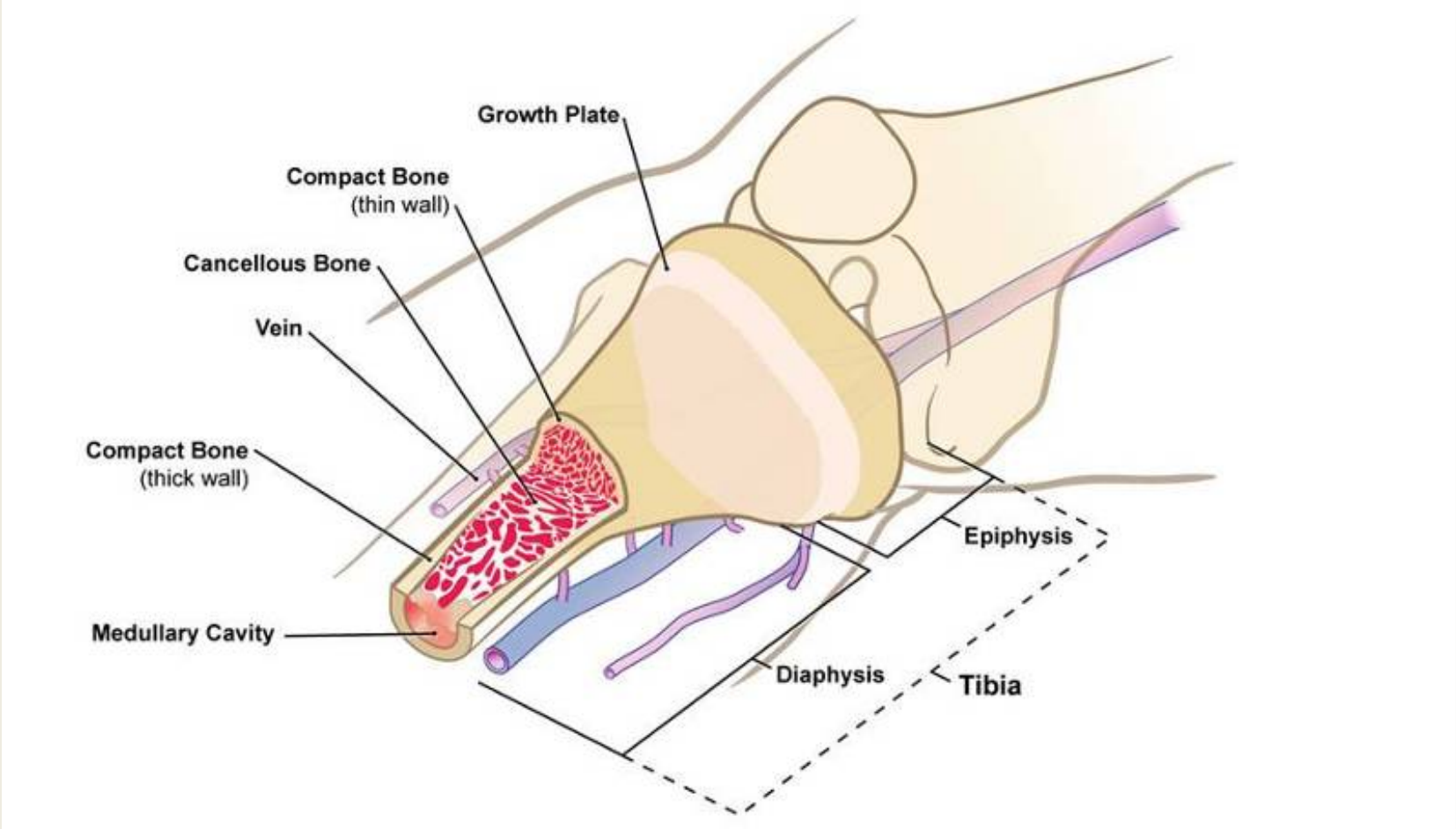
Insertion and Removal of Device

The insertion and removal procedure for the proximal humerus is identical as for other approved sites.



NOTE ON STABILIZATION: *Movement of the arm above the plane of the shoulder can result in needle dislodgement from impingement of the needle set on the acromion process of the scapula. Following insertion of a humeral IO device, the patient's arm should be immobilized to prevent movement above the level of the shoulders. In most cases restraint in the adducted position is preferred. The IO device itself should also be secured to prevent accidental dislodgement (the EZ-Stabilizer is designed for this purpose).*

Pediatric Anatomical Overview



Identifying the EZ-IO PD insertion site

If the Tibial Tuberosity CANNOT be palpated the insertion site is two finger widths below the Patella (and then) medial along the flat aspect of the Tibia

The Tibial Tuberosity can be difficult or impossible to palpate on younger patients

Identifying the EZ-IO PD insertion site

If the Tibial Tuberosity CAN be palpated the insertion site is one finger width below the Tuberosity (and then) medial along the flat aspect of the Tibia

As patients mature the Tibial Tuberosity becomes easier to identify

Indications for intraosseous access

- Cardiac Arrest
- Respiratory Compromise
  - Need for immediate rapid sequence induction
- Hemodynamic Instability
  - Mass Casualty Situations
  - Trauma Resuscitations
- Bridge to Central Line
  - Allowing Controlled Placement
- Altered Level of Consciousness
- Difficult IV Placement

Intraosseous Access = Immediate Vascular Access

Indications for Intraosseous Access

- **Patients with poor peripheral access**
  - Dialysis Patients
  - Sickle Cell Patients
  - Obese Patients
  - Mass casualty incidents
  - Congestive Heart Failure
  - Oncology Patients
  - IV Drug Abuse
  - Dehydration (especially pediatrics)
  - Diabetic Patient (DKA or hypoglycemia)

AHA, ERC, ILCOR, NAEMSP Guidelines

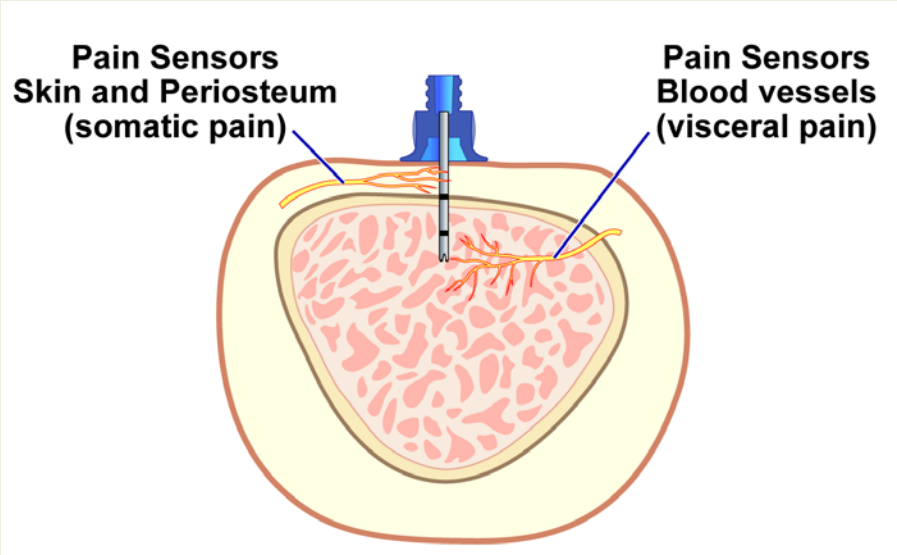
- IO should be considered early in vascular access emergencies
  - Adults - 2 peripheral IV attempts Progress to IO
  - Pediatrics - 1st line of choice
- ET tube is no longer recommended for drug delivery
- Central lines are discouraged
  - Approximately 5 million central venous catheters placed each year in US
  - Central line placement causes unnecessary drug delivery delay during resuscitation
  - CDC report indicates 9% infection rate with central lines in US

ILCOR is comprised of seven formal members – American Hear Association, European Resuscitation Council, Heart and Stroke Foundation of Canada, Australian and New Zealand Committees on Resuscitation, Resuscitation Councils of Southern Africa, and the InterAmerican Heart Foundation and Resuscitation Council of Asia

What About Infections With IO

- 20 + year history in pediatrics with Cook/Jamshidi needles sets
  - Overall infection rate is 0.6%
  - Cases of osteomylitis occurred when catheter was left in place for > 72 hours
  - Newer IO devices cause less bone trauma
- EZ-IO database
  - Contains 2000+ insertions with no local infections or osteomylitis
  - Estimate of 80,000+ insertions with no local infections or osteomylitis

Intraosseous access: is it painful?



- IO <u>insertion pain</u> is equivalent to a peripheral IV
- IO <u>infusion pain</u> can be severe but is significantly moderated by the administration of 20 – 40 mg Lidocaine for patients > 39kgs and 0.5mg/kg for patients 39kgs or less via the IO route (*2% preservative free Lidocaine is recommended)

Pressure and Flow Rates

- With pressure, IO flow rates are similar to IV
  - Tibial relates to a 18 gauge catheter
  - Humeral relates to a 16 gauge catheter
- Flow rates for infusions given through an IO with a 300 mm pressure infuser
  - 3 – 6 liters/hour of saline
  - Unit of blood in approximately 15 - 30 minutes
- Syringe bolus infusions can be completed in seconds
- Initial rapid 10 cc syringe bolus for patients > 39kgs and 5cc flush for patients ≤ 39kgs dramatically increases IO flow rates

NO FLUSH = NO FLOW

Contraindications

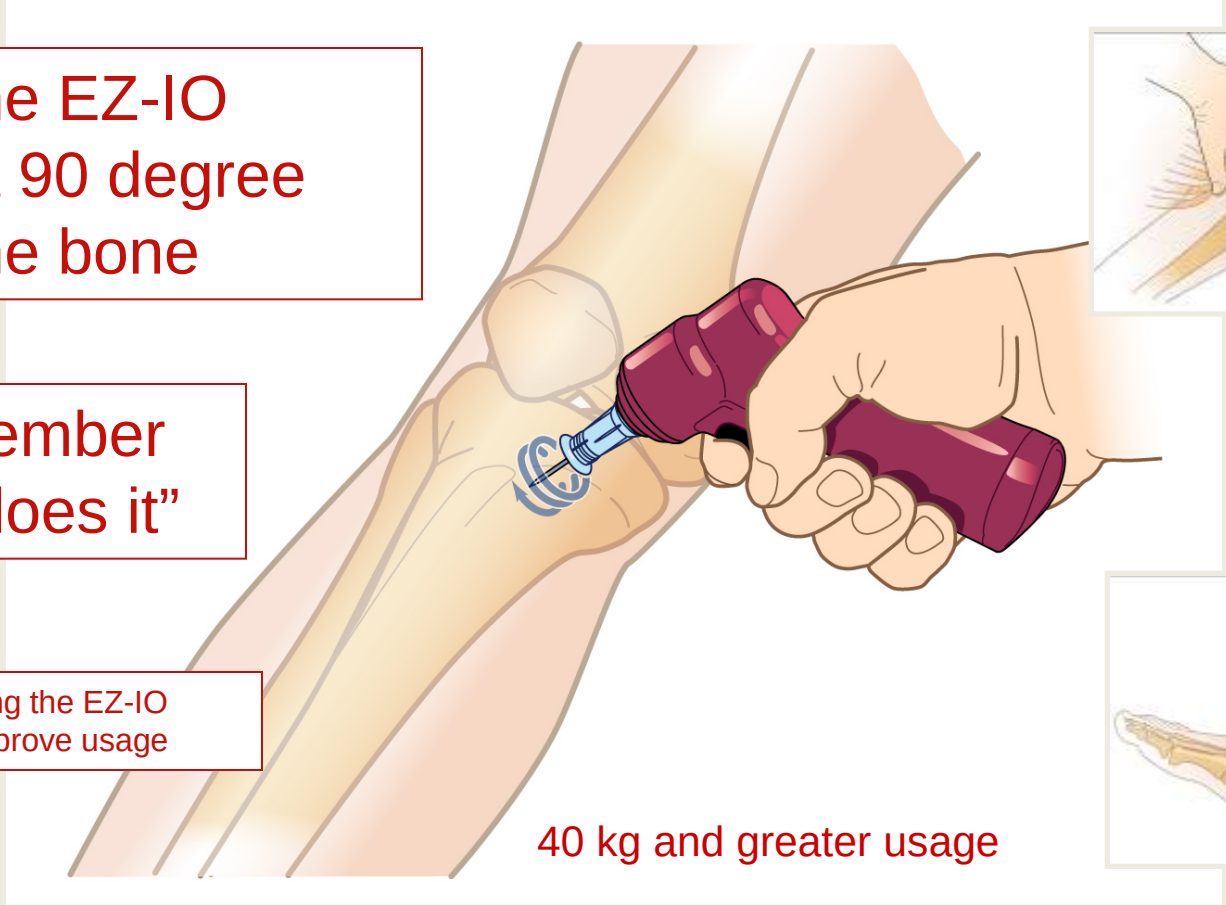
- Local Infection (at the insertion site)
- Fractures (to the bone selected for insertion)
- Prosthesis
- Recent (24 hours) IO in same extremity
- Absence of anatomical landmarks or excessive tissue

Insert AD needle set into appropriate site

Position the EZ-IO Driver at a 90 degree angle to the bone

Remember "EZ does it"

Lightly holding the EZ-IO driver will improve usage

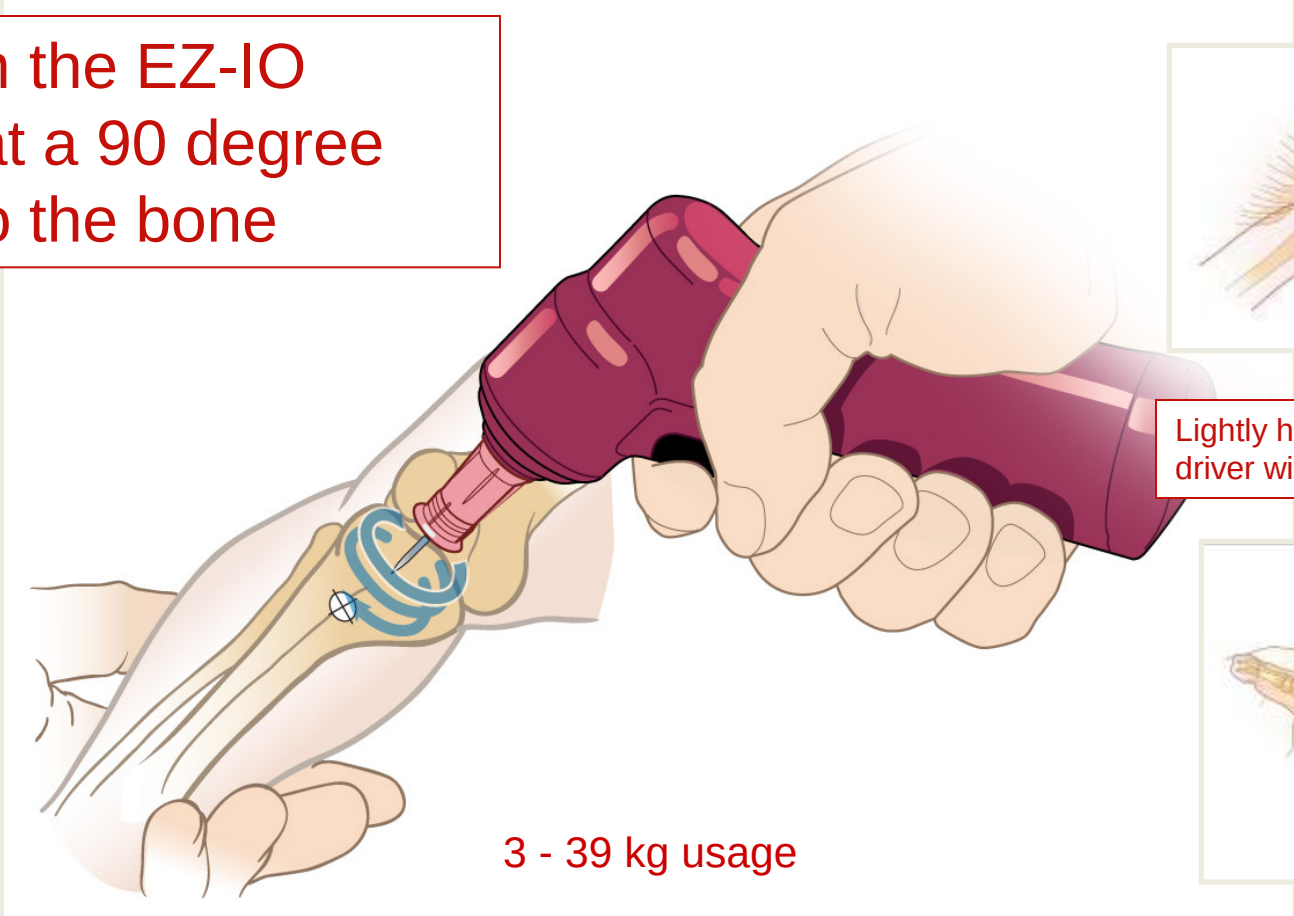


40 kg and greater usage

Don't force the needle set into position - "allow the driver to do the work"

Insert PD needle set into appropriate site

Position the EZ-IO Driver at a 90 degree angle to the bone



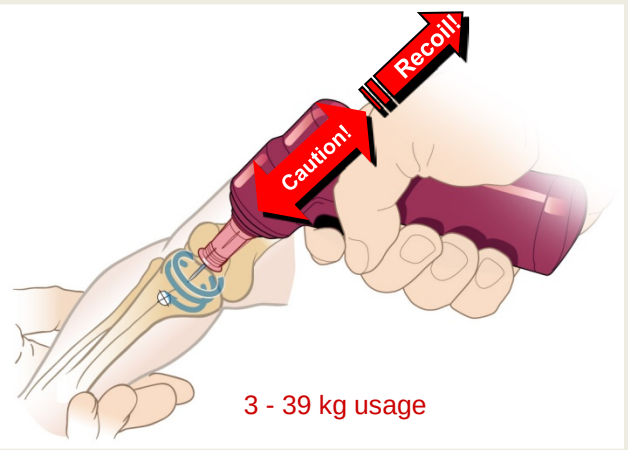
Lightly holding the EZ-IO driver will improve usage

Select needle set based on patient size & weight

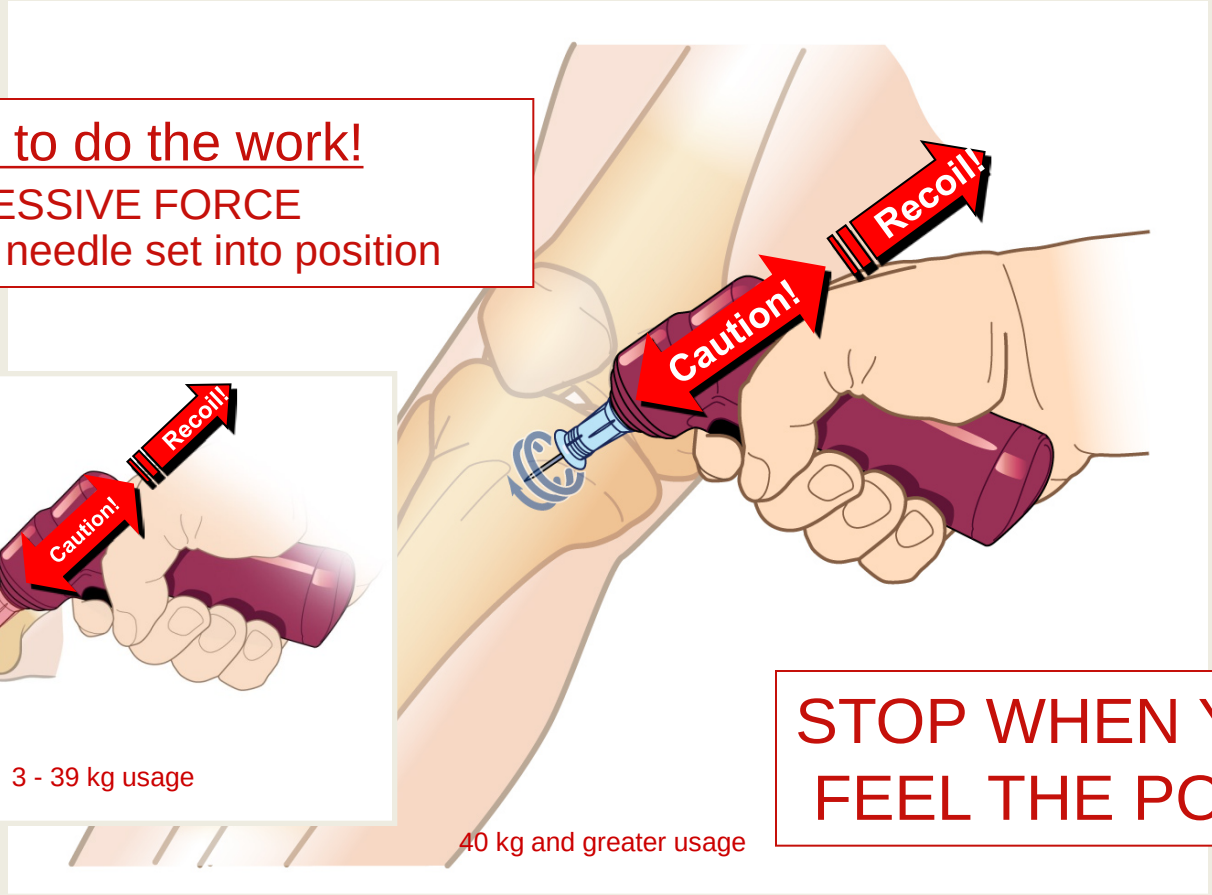
Important needle set insertion tip

<u>Allow driver to do the work!</u>

DO NOT EXCESSIVE FORCE
Gently GUIDE needle set into position



3 - 39 kg usage

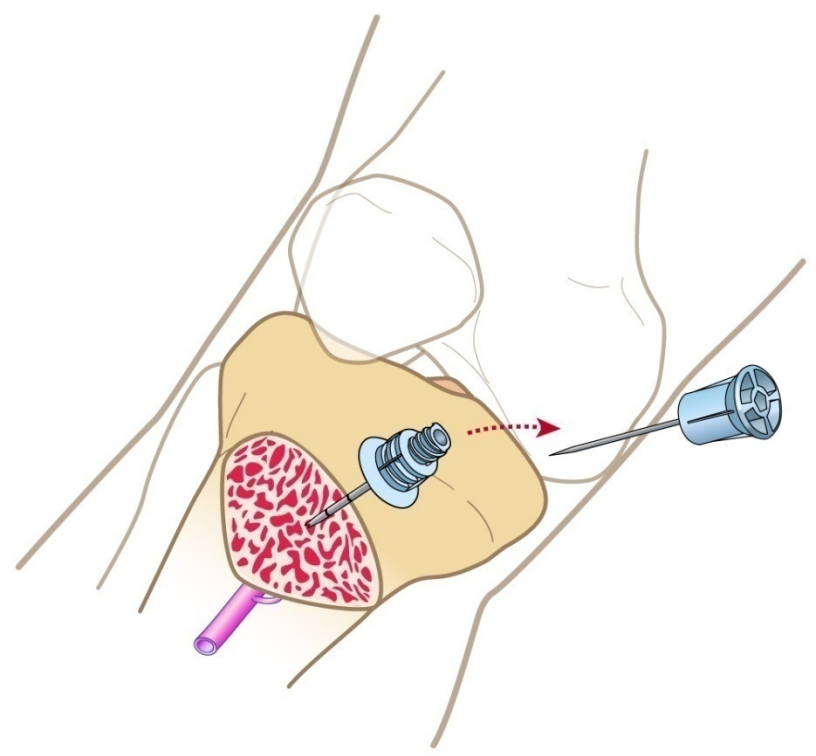


40 kg and greater usage

STOP WHEN YOU
FEEL THE POP

User induced recoil may lead to needle set dislodgement or extravasation

Remove stylet and confirm placement

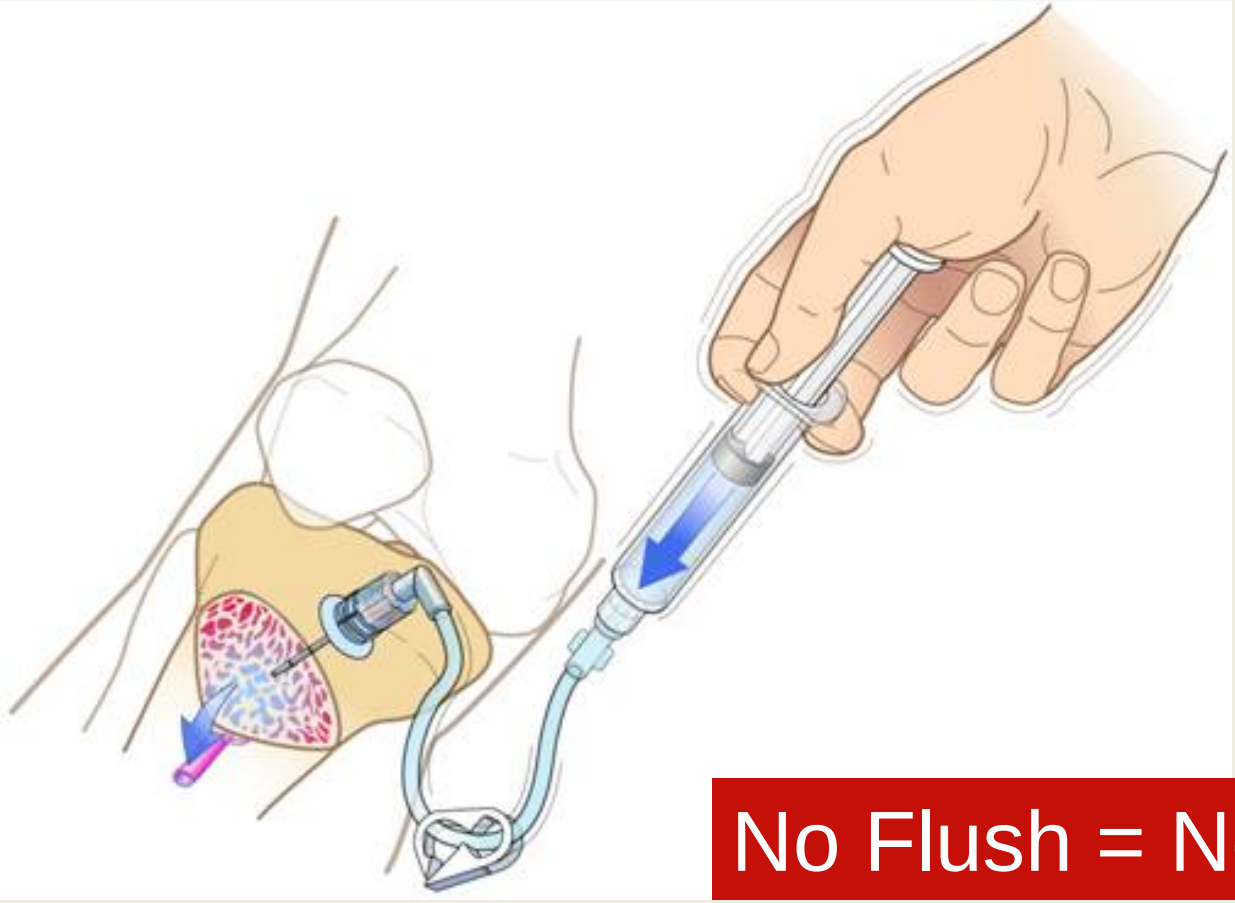


Confirm placement by noting

- Blood at the stylet tip
- Firmly seated catheter
- Blood in the catheter hub
- Aspiration of blood
- Fluids flow without difficulty
- Pharmacologic effects

Monitor the insertion site and distal extremity for signs of extravasation

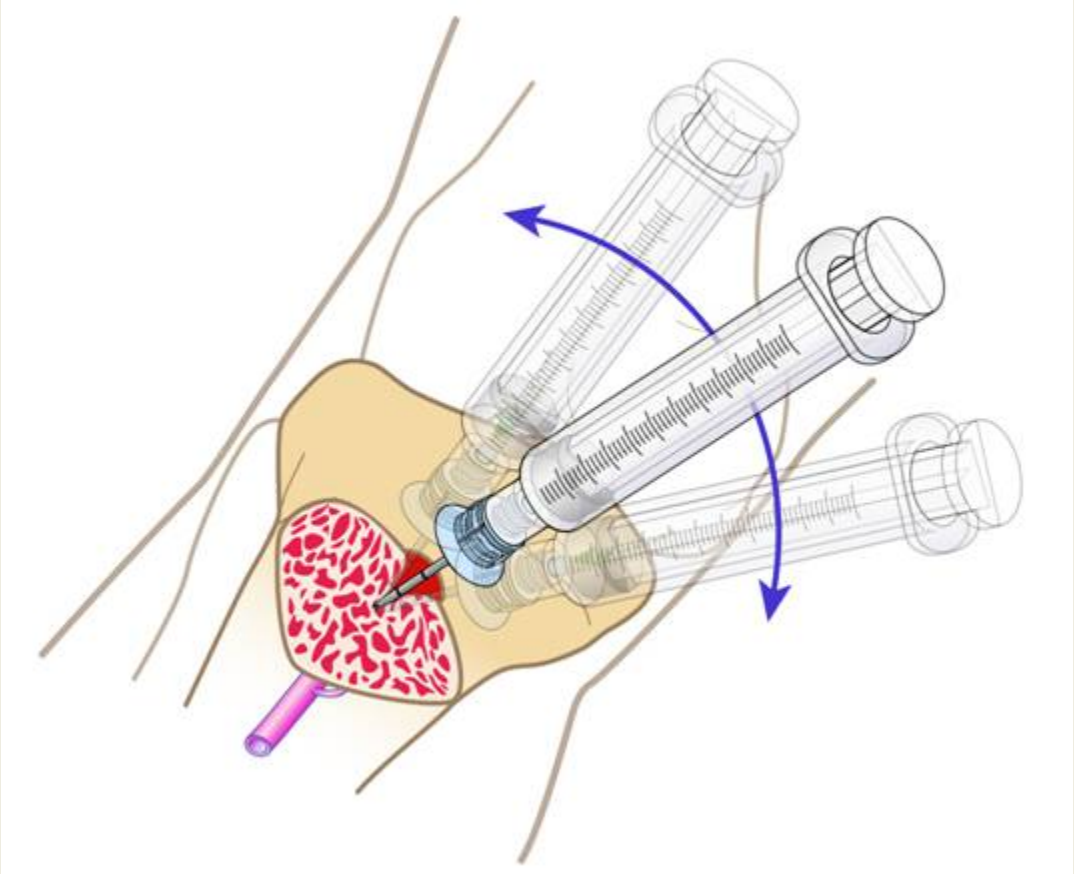
Syringe flush catheter



No Flush = No Flow

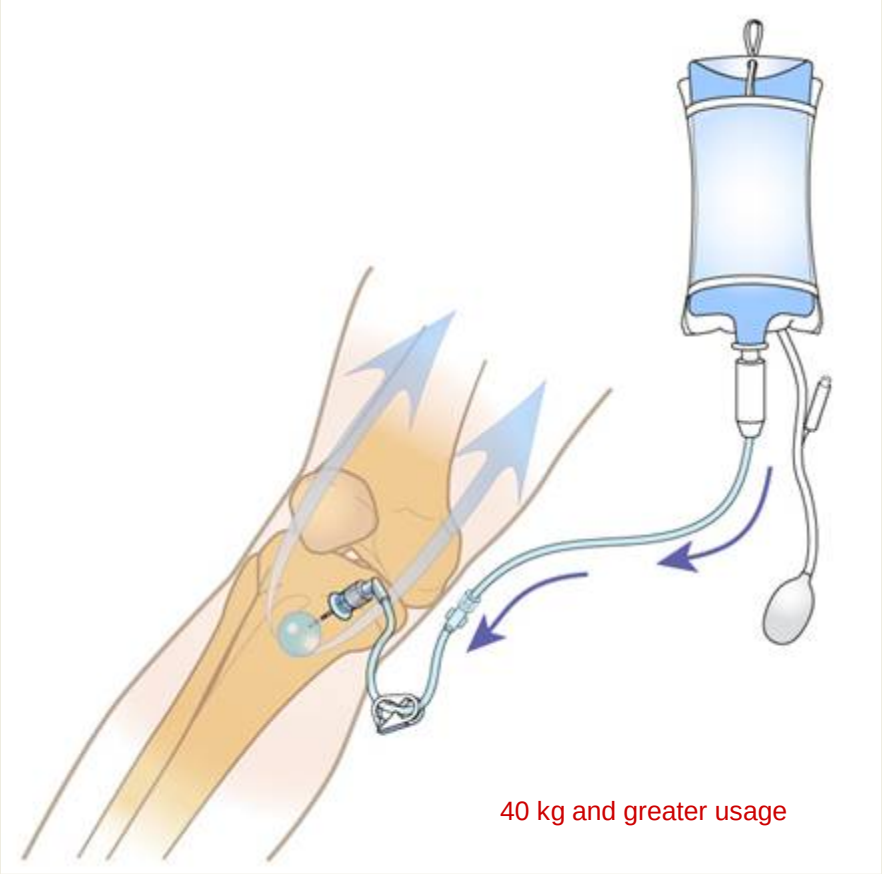
Syringe flush the catheter with 10 ml (5 ml for PD) of a sterile solution

Avoid rocking the EZ-IO catheter



Use the EZ-Connect supplied with the needle set!

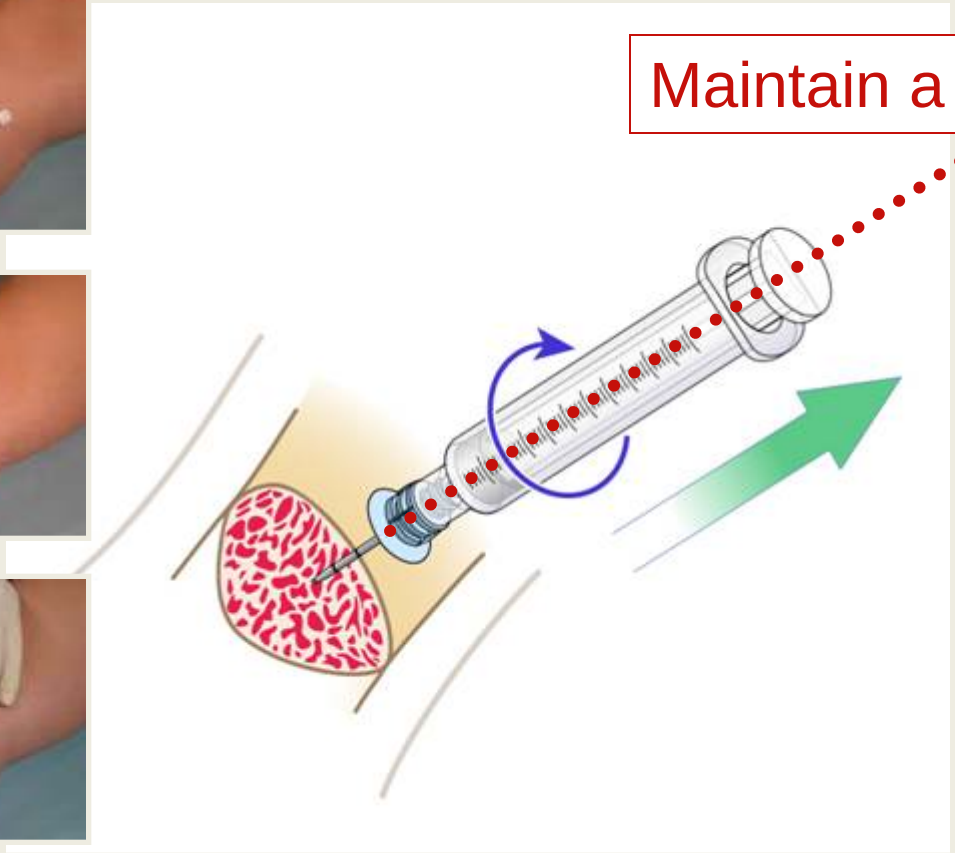
Begin infusion with pressure



Regulate fluid delivery for pediatric patients

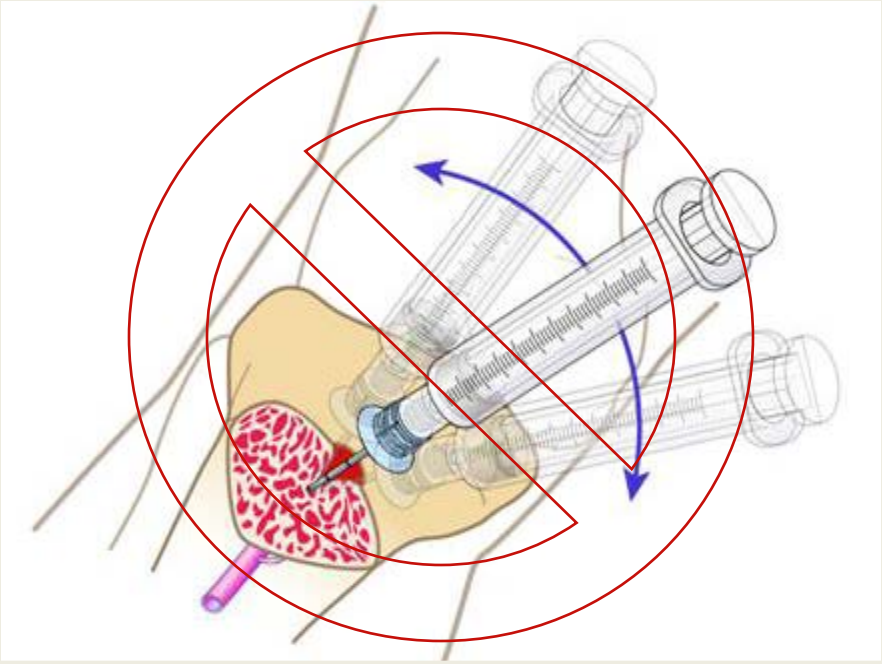
A pressure bag, infusion pump or syringe will improve the flow rates

EZ-IO Removal



Maintain 90 degree angle, Rotate clockwise and gently Pull

Possible Complications



Rocking, Bending or inadvertently Striking the catheter may cause it to break

If breakage occurs Grasp the exposed catheter with a hemostat – rotate and pull

Questions?